Requirement Analysis Phase

Customer Journey Map

Date	28 October 2026
Team ID	Ujjawal Rathod (Individual Capstone)
Project Name	Electric Vehicle Market Performance & Insights
Maximum Marks	2 Marks

Customer Journey Map: Maps out the stakeholder experience stage-by-stage across our three EV personas (Emily Johnson, David Chen, and Dr. Sarah Williams), capturing their actions, touchpoints, thoughts, feelings, process ownership, and improvement opportunities.

Phase of Journey	Stage 1: Raw Data Exploration & Problem Identification	Stage 2: KPI & Visual Discovery (9 Tableau Worksheets)	Stage 3: Interactive Dashboard Cross-Filtering	Stage 4: Executive Storytelling & Web Portal Access
Actions (What does the customer do?)	Attempts to inspect `ev_market_train.csv` (200k rows) and `ev_market_test.csv` (100k rows) across 37 columns to find sales, range, and competitor trends.	Explores individual visual sheets (V1–V9) covering category sales, Top 10 manufacturers, model year trends, battery tech share, range, map, ratings, and heatmap.	Uses global filters (Market Region, Battery Type, Model Year) and click-to-filter actions on the Map (V7), Battery Donut (V4), and Category Bars (V1).	Navigates the 4-point Tableau Story (Overview, Emily, David, Dr. Sarah) embedded inside the deployed Flask web application on Vercel.
Touchpoint (What part of the service do they interact with?)	Raw CSV files and Tableau Data Source canvas (300,000-row Union).	Tableau Public Worksheets V1 to V9 with custom FIXED LOD calculations and zero-anchored axes.	EV Market Dashboard (3×3 tiled grid layout, 1400×1500 px) on Tableau Public.	4-Point Tableau Story & Live Flask Web App (`index.html` + `app.py` hosted on Vercel).
Customer Thought (What is the customer thinking?)	"There are 300,000 rows, 200 manufacturers, and missing values in sales (~1%) and ratings (~2%)—how do we extract reliable answers quickly?"	"Two-Wheeler Scooter leads sales, Ampere Works holds ~0.59% share in a fragmented market, and range depends on category (550 km vs. 105 km), not battery type (~324 km)."	"Clicking a country or battery slice updates all 9 charts in 1–2 seconds, letting me compare regional product positioning immediately."	"Having dedicated persona views and a single Vercel web link makes presenting to the board effortless."
Customer Feeling (What is the	**Overwhelmed, frustrated,** and hesitant to make high-stakes decisions from raw spreadsheets.	**Curious, engaged,** and reassured by accurate FIXED	**Empowered, in control,** and analytical while testing regional	**Confident, satisfied,** and ready to execute data-

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customer feeling?)		LOD market shares and honest axis scaling.	and technology scenarios.	backed strategic decisions.
Process Ownership **(Who is in the lead on this?)**	Data Analyst / BI Developer (Ujjawal Rathod) — Data ingestion, Union setup, and schema validation.	BI Developer (Ujjawal Rathod) — Designing V1–V9 visuals, FIXED LOD formulas, and reference lines.	BI Developer & Stakeholders (Emily Johnson, David Chen, Dr. Sarah Williams) — Interactive filtering.	Web/BI Developer (Ujjawal Rathod) & Executive Stakeholders — Story presentation and Vercel deployment.
Opportunities **(How can we improve this stage?)**	Automate CSV unioning and document null handling (Tableau automatically excludes nulls in SUM/AVG).	Use `{FIXED : SUM([Annual Sales Units])}` for true market share and anchor V6/V8 axes at 0 to prevent distortion.	Remove duplicate gradient legends and cell label clutter (###) in the V9 heatmap for clean readability.	Embed both the Dashboard and Story in a responsive Flask web portal deployed on Vercel for 24/7 browser access.